Cloud Computing

Delivery of computing services over the internet

SAY GOODBYE TO PHYSICAL STORAGE

Store your files in the cloud and access them anytime anywhere

YOUR COMPUTER IN THE CLOUD

Run Software,Test apps or Host Systems-without needing physical hardware.

SMART DATA LIVES IN THE CLOUD

Store,Manage and access data quickly and securely from anywhere

CONNECTIONS THAT POWER THE CLOUD

Cloud networking keeps your data flowing smoothly and securely

Kavindya Mallawa

Shared Responsibility Model

Cloud security split between provider and customer

YOU'RE RESPONSIBLE FOR

- Your data and information
- Account and Identity
- Devices connecting to the cloud

CLOUD PROVIDER IS RESPONSIBLE FOR

- Physical data center security
- Physical Network
- Host infrastructure

THE CLOUD PROVIDER SECURES THE FOUNDATION, BUT YOU CONTROL WHAT'S BUILT ON TOP

Kavindya Mallawa

Cloud Models

how cloud services are deployed and consumed

PRIVATE CLOUD

Cloud environment dedicated to a single organization

- More control, but higher cost
- Ideal for sensitive workloads

PUBLIC CLOUD

offers shared resources managed by third-party providers,accessible to anyone.

- Lower cost and high scalability
- Fast to deploy and easy to use

HYBRID CLOUD

Combines public & private to provide organizations with flexibility,scalability and enhanced security.

- Customize deployment locations
- Scale workloads when needed

Kavindya Mallawa

Consumption-Based Model

Pay Only for What You Use

Cloud computing uses a pay-as-you-go model where you rent IT resources and pay only for what you use.

CAPITAL EXPENDITURE

One-time costs like building datacenters or buying servers

OPERATIONAL EXPENDITURE

Ongoing costs like renting cloud services or paying monthly bills

Benefits

- No upfront investment needed
- Scale resources up or down instantly
- Avoid wasted capacity or performance issues
- Cloud provider handles maintenance and infrastructure

Kavindya Mallawa

Benefits of Using Cloud Services

HIGH AVAILABILITY

Ability to keep services running for extended period of time with little downtime

SCALABILITY

Ability of system to handle growth of users and work

AGILITY

Ability to react quickly & focuses on speed & ease of scaling resources

ELASTICITY

Automatically grow and shrink based on app demand

RELIABILITY

Ability of a system recover failures and continue to function

Kavindya Mallawa

Cloud Service Type

Infrastructure as a service (IaaS)

lets you rent cloud-based hardware with full control over software and configurations, while the provider manages the physical infrastructure and networking

Platform as a service (PaaS)

Provides a ready-to-use development environment where the provider handles infrastructure,OS and tools,letting you focus on building and deploying applications.

Software as a service (SaaS)

Delivers fully functional applications over the internet,allowing users to access and use software without managing infrastructure or maintenance.

Kavindya Mallawa